



Karbonetiq

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Karbonetiq Inc.

Company or organization location (we welcome applicants from anywhere in the world)

Santa Barbara, CA

Name(s) of primary point(s) of contact for this application

Luke Rondel - VP Commercial
Dr. Michael Wyrsta - CEO

Brief company or organization description <20 words

Founded in 2022, Karbonetiq valorizes industrial byproducts through mineralization CDR to support industrial decarbonization.

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

Gigaton Potential

Karbonetiq's CDR enhances mineralization of alkaline industrial byproducts (slag, lime, ashes) to capture and store atmospheric CO₂. Our engineered system maximizes gas-solid reaction surface

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

area and minimizes energy inputs to provide the greatest net-removal by any performance metric (cost per ton of CDR, CDR per unit of energy, etc.).

Gigatons of legacy byproducts reside in landfills and waste ponds, and essential industries (steel manufacturing, power generation, etc.) continuously produce vast quantities of new alkaline byproducts. RMI estimates these materials may result in ~1.3 gigatons of CDR by 2050, while other estimates place the potential in hundreds of megatons of direct removals and gigatons of indirect reductions.²

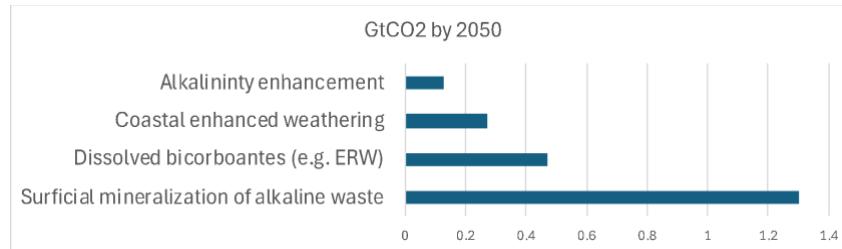


Figure 1. According to RMI, surficial mineralization of alkaline waste has the potential to sequester >1 Gt of CO₂ by 2050.

Carbonation is the reaction of CO₂ with alkaline materials, typically rich in calcium or magnesium, that creates solid carbonate minerals. Calcium Carbonate (CaCO₃), commonly referred to as calcite or limestone, is a geologically stable mineral that endures in natural conditions for millions of years, during which time carbon captured through mineralization is locked in a crystalline lattice.

Steelmaking slag, a byproduct of metal production, contains calcium silicates and oxides. When exposed to air, these components spontaneously react with atmospheric CO₂, often aided by water vapor, to form calcium carbonate *via* the simplified chemical reaction:



While slag has the chemical potential for CDR, common ubiquitous storage techniques lead to 'passivation' where the material forms a carbonate crust that arrests CO₂ absorption at very low levels. Karbonetiq has analyzed hundreds of slag samples, some many decades old, that demonstrate an average carbonation rate of less than 5% of the material CDR potential. Karbonetiq's goal is efficient exploitation of the remaining 95%.

Operational and Technological Innovation

Karbonetiq's team leveraged decades of industrial logistics experience to develop a mineralization CDR approach that valorizes underutilized material and reduces operators' waste liability. Our approach provides additional co-benefits through enhanced metal recovery (a key component of decarbonized steelmaking) and improved material stability for downstream uses such as aggregate.

Operationally, our approach minimizes barriers to impact by integrating with existing industrial practices, working with developed skillsets, following established permitting frameworks, and utilizing installed capital. Our projects are constrained to existing industrialized sites to reduce transport costs, secure operations, and mitigate social and environmental impacts. They also seek to minimize power demands, as electrification of primary industrial operations is already greatly increasing the load on the grid. Finally, our modular approach helps manage Capex requirements and facilitates scaling production to meet demand for CDR credits.

² Pan, SY., Chen, YH., Fan, LS. et al. CO₂ mineralization and utilization by alkaline solid wastes for potential carbon reduction. Nat Sustain 3, 399–405 (2020). <https://doi.org/10.1038/s41893-020-0486-9>

Karbonetiq CDR also includes an ever-expanding suite of proprietary technologies, including the Karbonator and TerraSeq™ MRV, that extract and precisely-measure the full carbonation potential of alkaline byproducts.

A Karbonator is a scaffolding system that maximizes gas-solid reaction surface area of granular material through enhanced surface area, improved airflow, and structural stability. In laboratory tests, crushing slag to expose new contact surfaces increases the carbonation potential approximately 20 times baseline, as shown in Figure 1 below. Karbonator enlarges the effective contact area of the activated slag, accelerating the carbonation rate by 34%. The combination of crushing and the Karbonator elevates the total carbonation rate of slag over a specified period. Compared to other passive aeration techniques, the Karbonator reduces land requirements by half. This doubles the CDR potential for projects with limited space and decreases the need for material transport. In aggregate, these improvements enable the production of CDR today at economically viable rates, and they provide what is likely the fastest and lowest risk path to achieving sub-\$100 per ton CDR by 2030.

The first step in Karbonetiq CDR is to employ low-energy crushing and screening to decrease particle size and expose fresh mineral surfaces to CO₂-containing air. The process then proceeds to the Karbonator. In addition to outperforming traditional aeration by more than 30%, the Karbonator further stands out from other approaches by requiring no operational power, pressurized airflow, or high-temperature processes, simplifying operations, and keeping costs low.

Karbonator improves the effective surface-to-volume ratio compared to unsupported piles of identical material by an estimated 1.34×. This geometric arrangement is mirrored by a measurable increase in carbonate content of ~1.3× compared to that of a cone pile under the same conditions. During lab testing, in-pile CO₂ concentration data indicates continuous carbonation activity in Karbonator, confirmed by a “traveling baseline” phenomenon (i.e., local CO₂ concentration fluctuations were influenced by active mineralization). Enhanced carbonation in the Karbonator is further supported by Thermogravimetric analysis and mass spectrometry indicating more CO₂ release from fully carbonated slag than fresh slag, and X-ray diffraction (XRD) indicating calcium hydroxide converted mainly to calcium carbonate in the Karbonator system.

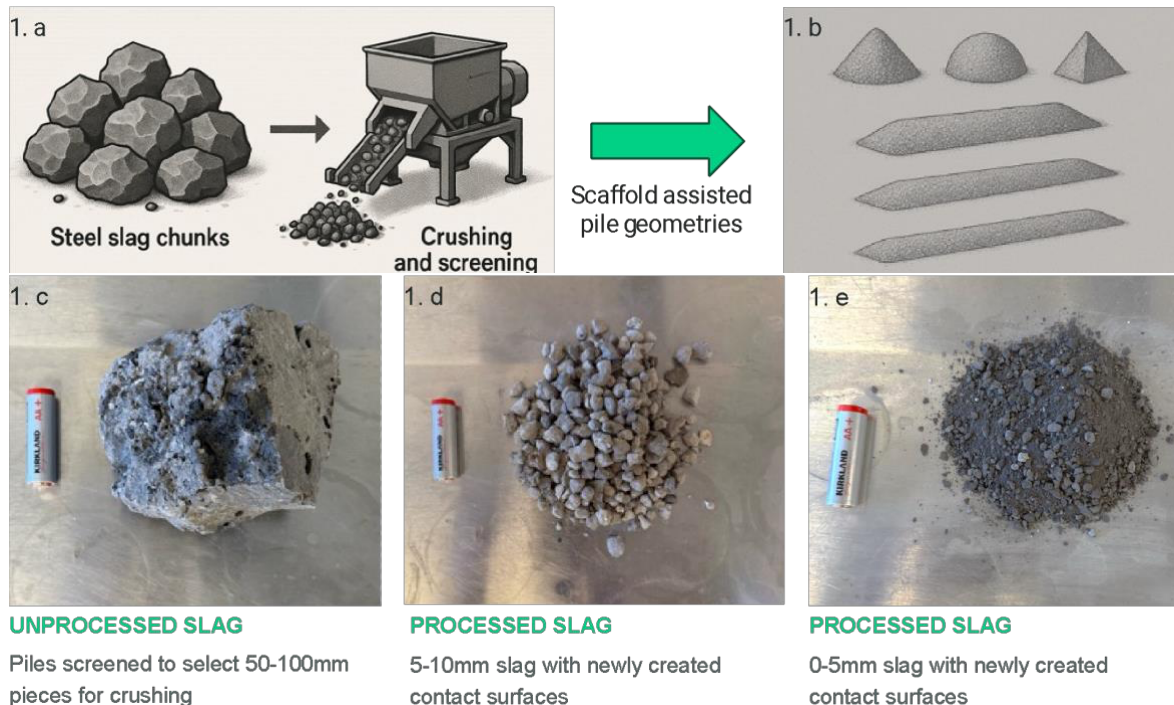


Figure 1. a. shows oversized, passivated steel slag being activated via crushing and screening; b. shows the scaffold-assisted (proprietary) layout of crushed and screened slag, which enhances the surface area and contact area and maximizes land utilization per ton of deployed slag. Panels c.-e. show the initial sizing of unprocessed slag relative to a standard aa battery, subsequent crushing and the sizing material produced from crushing (d. and e.).

TerraSeq is a proprietary breakthrough in mineralization CDR quantification. Unlike conventional approaches, TerraSeq triangulates multiple diverse data streams to create a high-fidelity picture of total CO₂ sequestered. As further described in later sections data streams include Smart Sampling Algorithms, Multimodal Material Analysis, and CO₂ Flux Measurement.

Differentiation

The Karbonator approach of operating at ambient conditions with passive aeration accelerates maximal carbonation of feedstock material without relying on additional energy inputs or complex reactor vessels. This significantly reduces costs and complexities, and their associated risks, compared to many more complex CDR approaches that use fans, heat, or electricity for sorbent regeneration.

The gas > solid capture and storage approach ensures exceptional stability: instead of using storage like compressed CO₂ tanks or geological injection—which can face leakage and monitoring challenges—this method converts CO₂ directly into solid carbonate minerals, effectively eliminating re-release risks.

Karbonetiq's durable mineralization relies on readily available steel slag and other waste, transforming an abundant industrial byproduct into a vital resource for carbon removal.

Passive aeration typically requires significant land usage, and here Karbonator further differentiates itself from other approaches by minimizing material transport and reducing the required area for carbonation by at least 2x. As the system reaches optimal carbonation in a matter of weeks, material can then be moved to more concentrated storage, further reducing land-use requirements. The design can be scaled modularly to nearly any field size and geometry, adapting to various land constraints with versatile spacing or stacked configurations, ensuring best-in-class economic and operational viability.

Looking Forward

Karbonetiq is poised to activate a vast stockpile of latent CDR potential with innovative technology and a practical approach informed by decades of operational experience. We are initiating 2 projects in Q2 2025 adjacent steelmaking facilities in Quebec, Canada and Ontario, Canada. We have designed a low-cost, CDR system that can scale with efforts to reduce greenhouse gas emissions, and that is precisely measurable through robust monitoring, reporting, and verification (MRV). This system will be net-negative in 2025 and will scale beyond initial projects based on demand.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?³ What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

Global Ambitions

Karbonetiq is building a network of projects, co-located with ongoing industrial operations, that contribute to broader industrial decarbonization by valorizing waste streams. Our projects minimally disrupt primary activities, such as steel production, and build on generations of established knowledge and skillsets. Karbonetiq leads project development, implements our technology to

³ We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

generate and measure CDR, and manages revenue and financing. We work with longstanding partners to execute project work. CDR is waste management, and Karbonetiq is well positioned to grow the work to transformative, and climatically relevant, scales.

Siting and Scale

Our approach is best-suited for deployment at or near industrial sites whose work produces an alkaline byproduct. Steelmaking facilities are one suitable type, thousands of which have vast stores of legacy slag and continuously generate fresh slag. We prioritize sites for project development based on the following criteria:

- Potential for impact
 - Slag Availability: Large steelmaking facilities ensure a steady supply of feedstock.
 - Land Availability: Each Karbonator requires a relatively modest area, but we look for ample space to expand.
- Partnerships – We aim to co-develop projects with organizations that are committed to sustainability and contributing positively to the communities in which they are located.
- Regulatory Environment: Favorable regulations can expedite permitting, project development, and financing.
- Local Climate: While Karbonetiq CDR operates effectively across diverse climates, moderate conditions with sufficient humidity can further aid carbonation.

Scaling Strategy

We will launch early projects with a small number of Karbonators at each location, then expand to dozens of Karbonators capturing thousands of tons of CO₂ per year once optimal operating parameters are refined. Projects at full utilization will deploy hundreds or thousands of Karbonators capable of removing tens-of-thousands to millions of tons of CO₂ annually.

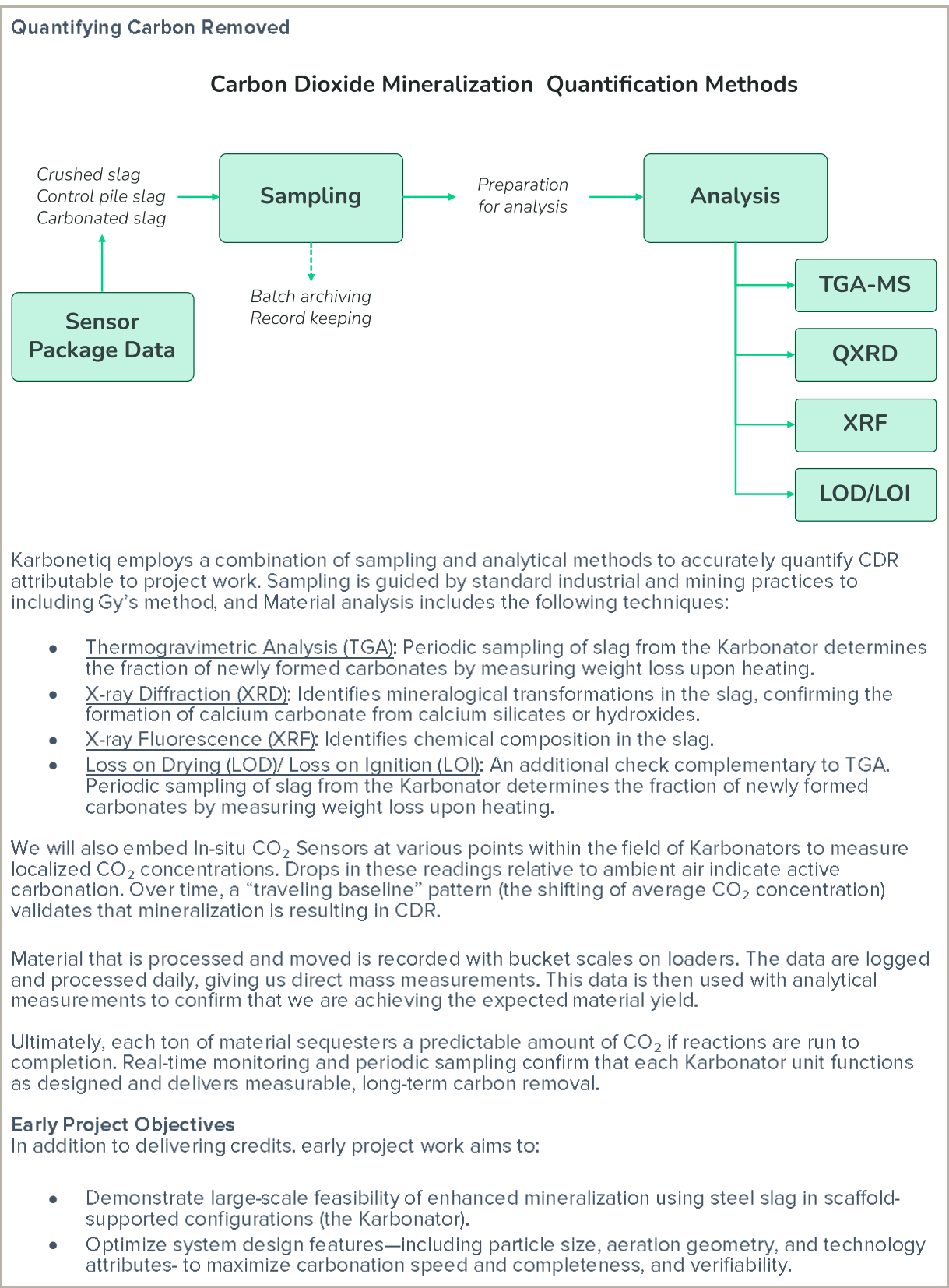
Initial Projects

Karbonetiq initiated operations at our pilot project - Contrecoeur CDR - in April 2025. Karbonetiq is the project proponent with full responsibility for project design, engineering, operations, and verification. We are partnered with Harsco Environmental to access land and feedstock and to operate the project. Harsco has been the trusted global leader in environmental services and innovative products for the steel-making industry for more than 150 years. Harsco has operated the site where our pilot will take place for many decades and processes ~1,000 tons of slag per day from surrounding steel production facilities, including ArcelorMittal Contrecoeur and Rio Tinto Fer et Titane. Harsco estimates ~1,250,000 tons of legacy slag material on site.

The first Karbonators are complete and are scheduled to arrive onsite in April. Our material-analysis lab in Santa Barbara is fully operational and has been characterizing material from this site since mid-2024, and we have successfully completed Puro.Earth's Pre-Assessment for this project. We will work with Puro to verify credits in Q3/Q4 2025.

We are also launching a project with Algoma Steel in Sault Ste. Marie, Ontario in Q2 2025. Algoma is an integrated steel producer that has been in operation since 1901. The facility generates enough slag to support 70,000 tons of CDR annually and includes an 800-acre storage yard with enough legacy slag to support up to 5 megatons of CDR. We chose Algoma for its scale potential and for the company's commitment to sustainability. The latter is exemplified by Algoma replacing their Basic Oxygen Furnace with an Electric Arc Furnace that will reduce the facility's emissions by ~70%. Karbonetiq is leading project development, with Algoma providing land and feedstock. We are partnering with Harsco Environmental to operate the project, utilizing insights from our pilot project and expanding our strategic partnership.

Both projects are "additional" as they are neither required by law nor is carbonation of slag a "common practice." Both projects are wholly dependent on carbon financing. Frontier's credit purchase will directly fund project development, MRV, and credit issuance. Minimal technology, regulatory, or stakeholder risk remains for either project; their scale and impact will scale to demand and access to developmental capital.



- Iterate on robust monitoring, reporting, and verification (MRV) protocols that accurately measure CO₂ mineralized and locked away in stable mineral form.
- Validate economically robust approaches to scaling, ensuring we operate on a trajectory to align with Frontier's pricing guidelines for carbon removal as we expand deployments.

Current Cost Breakdown

Cost considerations are paramount if CDR is to achieve broad economic feasibility, as emphasized in Frontier's guidelines. Below is a high-level breakdown of expense categories for the Karbonator system currently:

- Karbonator Fabrication, Installation and Financing (60%)
- Slag Procurement and Processing (30%)
- Monitoring, Reporting, and Verification (10%)

The cost per ton of CDR is influenced by local factors like slag availability, land access, yield, and labor rates. We project costs of ~\$325 per ton for early-stage, small-scale projects. Our plan for reducing costs through an optimized production process includes:

- Minimizing material transport - By situating Karbonators adjacent to steel production facilities, we drastically reduce material transport costs while ensuring continuous feedstock.
- Modular Construction - Standardizing the dimensions and components of each Karbonator module reduces fabrication times and simplifies maintenance and allows us to take advantage of volume discounts on materials and labor.
- Data-driven Process Optimization - We will continuously monitor carbonation rates using in-situ sensors and laboratory analyses and incorporate this data into the design process. This will allow us to refine elements such as particle size distribution, slag bed shape, and mesh porosity, thereby enhancing carbonation efficiency and minimizing the footprint needed for each ton of CO₂ removed. Karbonetiq has tested 245+ slags to date. Selecting the most reactive slags will play a key role in our cost reduction process.
- Economies of Scale - Operating as an industrial operation provides simple economies of scale that will provide cost savings as we grow via purchasing power, process electrification, etc.
- Efficient Financing - As a venture-funded startup operating at a small scale, financing is a significant contributor to cost per CDR credit. Prepayments, debt financing, and self-financing from excess cashflow will unlock significant savings that would be passed along to customers.

These measures will help us move toward Frontier's expected benchmarks. If we can drive final per-ton removal costs into the \$100/t range, our approach will stand among the most promising mineralization-based CDR solutions worldwide.

Looking Forward

We envision a future where widespread Karbonator deployment generates hundreds of millions of tons of durable CDR annually that directly supports broader industrial decarbonization. This aligns with Frontier's aim to support high-durability, scalable CDR that includes transformative co-benefits. By blending a simple yet innovative design with a commitment to data-driven optimization, the Karbonator is ready to meet the ambitious cost and scale criteria that climate action demands.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

Karbonetiq has engineered most risk out of our CDR, by designing our approach from industrial practices developed over generations and by repurposing existing processes and equipment. The scientific foundations of the approach are well established and present minimal additionality and

durability risk. We are also committed to engaging stakeholders to build strong support for our work and evolve our projects to best support community requirements.

Karbonetiq has allocated the necessary financial resources to cover all capital expenditures for the initial installation of technology and monitoring equipment and to cover operating expenses until projects can generate revenue. Comparatively little technology, regulatory, developmental, or stakeholder risk remains for this project; instead, the scale and pace of impact will be shaped almost entirely by demand for credits. In this light, Frontier's support will significantly impact our ability to launch a project of meaningful scale and impact.

2025 project operations will be our core technology's first field trial, and our project-execution team has developed the following list to track risks and apply mitigation strategies.

Technical Risks

- Slag Quality and Variability - Steel slag can differ significantly in composition depending on feedstocks and manufacturing processes. Some batches may have a higher calcium content (ideal for carbonation), while others could contain less calcium, reducing overall carbon sequestration per deployment.
Mitigation: Karbonetiq directly analyzes all feedstock batches for mineral composition and hazardous contaminants before use. Blending can be employed to ensure consistency.
- Carbonation Kinetics - While enhanced mineralization in slag is proven, reaction rates can slow down if the surface layers become passivated. This is especially relevant if the particle size is too large or the slag compacts into a dense mass.
Mitigation: Our Karbonator design addresses this by maximizing the surface-to-volume ratio and maintaining airflow channels. Regular monitoring of CO₂ uptake guides adjustments in slag grading (particle sizing) or pile geometry to sustain optimal reaction rates.

Project Execution Risks

- Construction and Logistics - Building multiple Karbonator units and ensuring a reliable slag supply may face issues such as labor shortages, bottlenecks in materials delivery, tariffs, trade wars, or local regulatory hurdles.
Mitigation: Detailed project scheduling and phased construction allow us to learn from smaller pilot deployments before scaling up. We also plan to source local materials to reduce lead times and manage strong relationships with local steel producers or suppliers.
- Scale-up Challenges - When expanded to hundreds or thousands of modules, an initially successful pilot or demonstration may encounter unexpected hurdles, such as land constraints or more stringent safety requirements.
Mitigation: Our system is inherently modular, meaning each Karbonator unit is self-contained. This reduces the complexity of scaling up; lessons learned at smaller scales (e.g., airflow patterns, slag throughput rates) can be directly applied to larger deployments with minimal redesign. Our operating Partner Harsco brings more than 150 years of experience and operational expertise navigating industrial sites around the world.

Measurement, Reporting, and Verification (MRV) Risks

- Data Accuracy and Integrity - Tracking CO₂ uptake requires laboratory-based analyses (e.g., thermogravimetric analysis, X-ray diffraction), and in-situ sensor measurements. If equipment malfunctions or sampling protocols are not rigorously followed, data may be unreliable.
Mitigation: We employ redundant sensors, standardized sampling procedures, and strict calibration protocols. Third-party audits can verify data quality, and automated alerts can signal deviations or equipment failures in real-time.

- Market and Credibility Risks - Carbon credits generated through mineralization must adhere to recognized MRV frameworks. If data are incomplete or verification standards are unclear, carbon credits may not be accepted on major exchanges.
Mitigation: We have engaged with established registration bodies early in project design to ensure our protocols align with recognized methodologies. Transparent documentation, open-source data (where appropriate), and peer-reviewed research all bolster project credibility.

Ecosystem and Environmental Risks

- Local Ecological Impact - Although slag is a byproduct, large-scale stacking or piling can affect local habitats, especially if new land must be cleared. Additionally, dust and runoff from slag piles could pose risks to nearby ecosystems.
Mitigation: We select sites that are already industrial or brownfield in nature, thereby minimizing habitat disruption. We also implement dust control measures and water drainage systems to protect surrounding soil and water sources and ensure compliance with environmental regulations.
- Heavy Metal Leaching - Some slags may contain trace metals, which could leach out over time, particularly if the pH changes.
Mitigation: Our proposed method for avoiding metal leaching is to select slags without heavy metals. Also, facilitating carbonation reduces the slag’s alkalinity, reducing the potential for metal mobilization. Where needed, periodic water testing around the site confirms that leachate is within acceptable limits. Should elevated levels appear, we can seal or further treat the affected slag to stabilize any problematic materials.

Financial and Market Risks

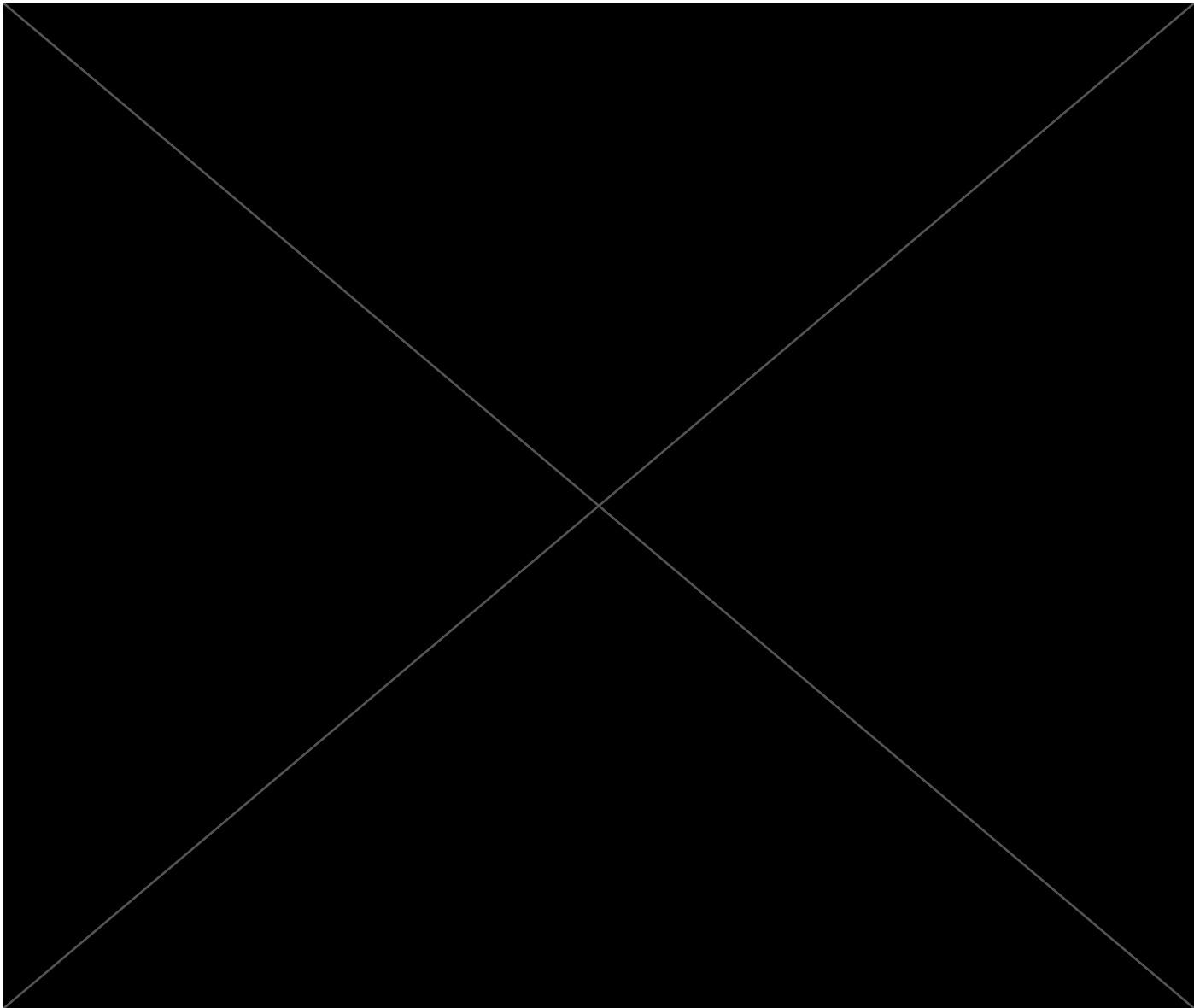
- Capital and Operating Costs - While the Karbonator system is designed to be cost-effective, fluctuations in steel slag availability or construction materials costs can inflate costs. Similarly, if labor rates spike or regulatory fees increase, per-ton costs may become unsustainable.
Mitigation: To reduce production cost volatility we are diversifying our slag supply, including to existing stockpiles and uncovering high-yield feedstocks, and negotiating long-term agreements with O&M contractors and fabricators. Modular design lowers initial capital expenses, enabling phased investments rather than a single large outlay.
- Carbon Price Volatility - If carbon markets weaken or demand for voluntary offsets falls, revenues from carbon credits may not fully cover operational and expansion costs. Current geopolitical volatility, especially related to climate projects, adds risk to the whole CDR industry. Mitigation: We plan to forge long-term purchase agreements with organizations committed to stable or rising carbon prices. Diversifying our funding sources, such as grants and corporate sustainability initiatives, further buffers revenue risks.

By anticipating these risks and embedding well-thought-out mitigation measures from the outset, Karbonetiq aims to establish a reliable, scalable, and cost-effective CDR pathway. Our modular design, stringent MRV protocols, proactive community engagement, and adaptive management strategies collectively lay the groundwork for a resilient and sustainable operation.

- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	1,428
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Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	Q1 2026 - Q3 2026
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	\$321
Levelized price (\$/ton CO ₂) ⁴ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	\$350



⁴ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).